ARTIFICIAL INTELLIGENCE-I

OPEN-ENDED ASSIGNMENT

Individual Assignment | Total: 10 Marks

EPIDEMIC SIMULATOR X

Agent-Based Epidemic Simulation Using NetLogo

An Agent-Based Simulation of Disease Spread, Vaccination and Recovery

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1. Introduction

EPIDEMIC SIMULATOR X is an interactive agent-based model created in NetLogo to show how an infectious disease spreads through a community. Instead of treating the population as a single group, the simulation represents each person as a moving individual (agent). As agents move and interact in a shared space, the simulation visually and numerically tracks disease spread, vaccination, and recovery.

2. Objective of the Project

- Model a population as individual moving agents whose close contacts cause disease spread.
- Show four distinct health states—healthy, infected, recovered, and vaccinated—using clear colour coding.
- Allow users to adjust key epidemic parameters such as population size, infection probability, infection radius, recovery time, vaccination rate, movement speed, and social distancing.
- Track the outbreak in real time using live counters (monitors) and an epidemic curve graph.

3. Project Overview

At the start of a simulation run, a population of agents is placed in a bounded 2D world. Based on the selected settings, a set portion starts vaccinated, most start healthy (susceptible), and one or more agents start infected. As the clock runs, agents move around continuously. When an infected agent comes close to a healthy agent, the illness can spread. After a set recovery period, infected agents recover. Live monitors and a continuous plot record how the outbreak starts, peaks, and ends.

4. Model Entities and Health States

Each person in the simulation is represented as a circular agent moving inside the world grid. Every agent belongs to one of four health states, shown by its colour:

Colour in Interface	Health State Category	Meaning in the Model
Green	Healthy / Susceptible	Not currently infected; can catch the infection if an infected agent comes nearby.
Red	Infected	Currently carrying the infection and able to spread it to nearby healthy agents.
Cyan / Light Blue	Recovered	Previously infected and has completed the recovery period.
Purple	Vaccinated	Belongs to the vaccinated group set by the vaccination-rate slider.

5. Simulation Rules and Working Mechanism

- **Agent Movement:** Agents move across the grid at each step. The *movement-speed* slider controls how fast they travel and mix.
- **Proximity & Infection Radius:** Infection spreads locally. An infected (red) agent can only pass the disease to healthy (green) agents located within its *infection-radius*.
- **Infection Probability:** Being nearby does not guarantee infection every time. When a healthy agent is inside the infection radius, the *infection-probability* slider sets the percentage chance that transmission occurs.

- **Recovery:** Infected agents stay contagious for the duration set by *recovery-time*, after which they turn cyan/light blue (recovered).
- **Vaccination:** The *vaccination-rate* slider sets the vaccinated input for the population, shown as purple agents.
- **Social Distancing:** A *social-distancing* switch is available on the interface to turn distancing behaviour on or off.

6. Parameters and Controls

The table below lists all controls on the left panel of the NetLogo interface and their values in the captured demonstration run:

Control / Parameter	Interface Type	Captured Value	Purpose
START / STOP / INITIALIZE	Buttons	Interactive	Starts, pauses, or resets the simulation population.
social-distancing	Switch	Switch control	Enables or disables social distancing behaviour.
population-size	Slider	605	Total number of agents created in the world.
infection-probability	Slider	38	Chance of transmission when within the infection radius.
infection-radius	Slider	1.5	Distance around an infected agent where spread can occur.
recovery-time	Slider	43	Time steps required for an infected agent to recover.
vaccination-rate	Slider	15	Vaccination setting (102 vaccinated agents in the 605 population).
movement-speed	Slider	0.4 – 0.5	Speed of agent movement (0.5 in Capture 1; 0.4 in Captures 2 & 3).

7. User Interface and Implementation

The interface is divided into three clear sections:

- **Left Panel (Controls):** Contains the *STOP SIMULATION*, *START SIMULATION*, and *INITIALIZE* buttons, the *social-distancing* switch, and the six parameter sliders.
- **Middle Panel (Monitors & Graph):** Displays six live numerical counter boxes next to the *Epidemic Curve* plot.
- **Right Panel (Simulation World):** Shows the moving agents colour-coded by their current health state.

Execution Cycle: Clicking *INITIALIZE* creates the agents, sets initial states, and resets the graph. Clicking *START SIMULATION* runs a continuous loop that moves agents, checks nearby healthy agents for infection, updates recovery timers, and refreshes the monitors and plot at every time step.

8. Simulation Outputs and Monitors

- **VACCINATED, HEALTHY, INFECTED, RECOVERED:** Four live monitors showing the exact number of agents in each state at that moment.
- **TOTAL INFECTIONS:** Cumulative count of all agents that have been infected since the run started.
- **PEAK INFECTED:** Highest number of simultaneously infected agents reached during the run.
- **Epidemic Curve:** Plots the population counts (vertical axis, 0 to 2000) over *Simulation Time* (horizontal axis), showing how healthy numbers drop as infections rise and recover.

9. Experimental Observations

The three screenshots capture three stages of a run with *population-size* = 605, *infection-probability* = 38, *infection-radius* = 1.5, *recovery-time* = 43, *vaccination-rate* = 15, and *movement-speed* = 0.4–0.5:

Captured Simulation Stage	Vaccinated	Healthy	Infected	Recovered	Total Infections	Peak Infected
Screenshot 1: Initial / Early Stage	102	502	1	0	1	1
Screenshot 2: Active Outbreak Stage	102	319	153	31	184	154
Screenshot 3: Later / Post-Outbreak Stage	102	10	0	493	493	178

10. Results and Analysis

- **Total Population Remains Constant (605):** At every stage, Vaccinated + Healthy + Infected + Recovered = 605 (Stage 1: 102 + 502 + 1 + 0 = 605; Stage 2: 102 + 319 + 153 + 31 = 605; Stage 3: 102 + 10 + 0 + 493 = 605).
- **Early to Active Outbreak:** The simulation starts with 1 infected agent and 502 healthy agents. In Stage 2, the infection spreads rapidly through close contact, rising to 153 active cases and 31 recovered cases (184 total infections, peak of 154).
- **End of Outbreak:** By Stage 3, active infections reach 0 and 493 agents have recovered after reaching a peak of 178 simultaneous infections. Ten healthy agents remain uninfected, and the 102 vaccinated agents stay constant throughout. These observations describe the captured progression in the provided screenshots rather than a multi-parameter comparison.

11. Conclusion

EPIDEMIC SIMULATOR X clearly demonstrates how individual agent movement and proximity rules create population-wide outbreak patterns. Using simple sliders, live monitors, and an epidemic curve, the model makes it easy to observe how transmission, recovery, vaccination, and distancing settings shape the course of an epidemic.

12. AI Tools Used and Disclosure

> **Tool Used:** ChatGPT (OpenAI).
> **Purpose:** Used for assistance with planning the simulation structure, suggesting and explaining NetLogo code logic, and refining the report wording and formatting. The simulation was run by the student to produce the three included screenshots. The student is prepared to explain the implemented rules and procedures during evaluation.

13. Simulation Screenshots

The three original NetLogo screenshots below show the early, active, and post-outbreak stages of the simulation.

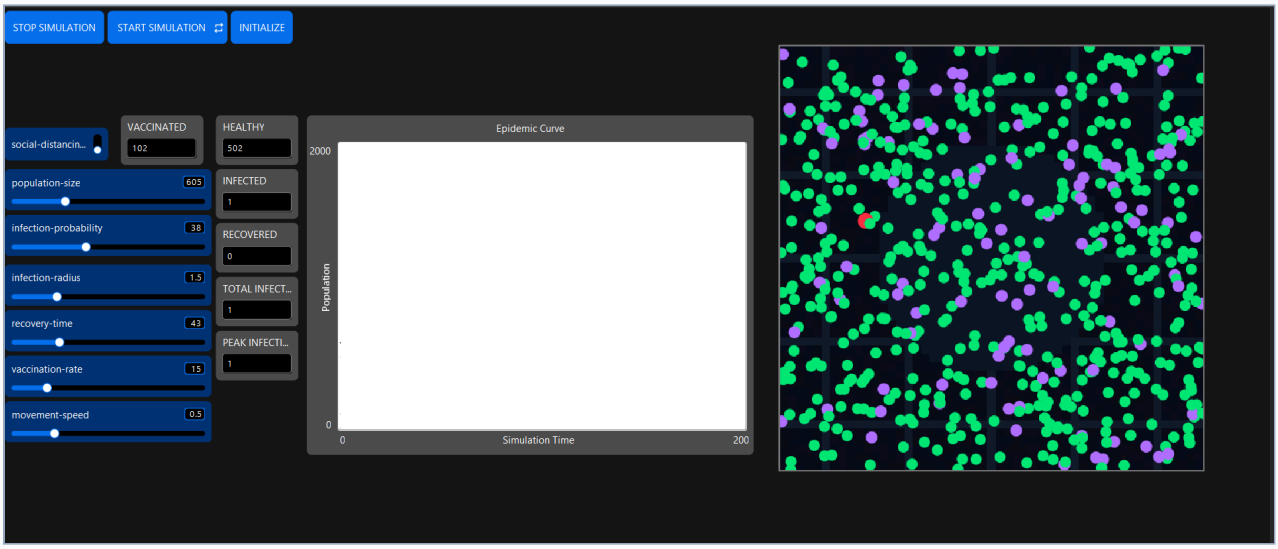


Figure 1: Initial / Early Stage of the Simulation. Captured immediately after initialization (*population-size* = 605, *infection-probability* = 38, *infection-radius* = 1.5, *recovery-time* = 43, *vaccination-rate* = 15, *movement-speed* = 0.5). The monitors record **102 Vaccinated**, **502 Healthy**, **1 Infected**, **0 Recovered**, **1 Total Infection**, and **1 Peak Infected**. In the world view on the right, a single red infected agent is visible on the middle-left amid green healthy and purple vaccinated agents.

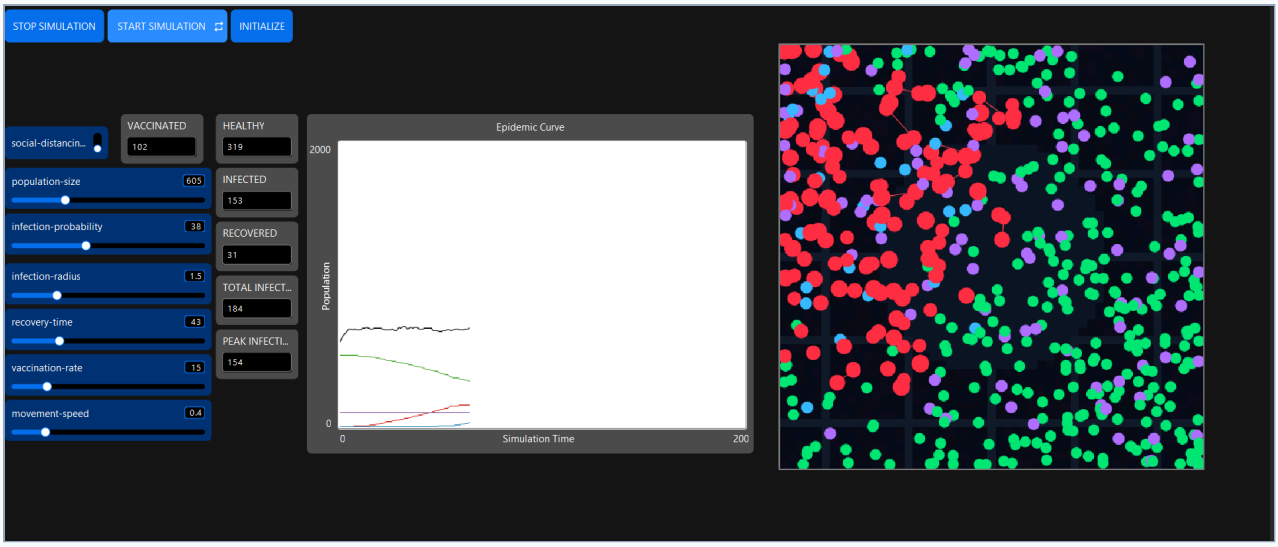


Figure 2: Active Outbreak Stage of the Simulation. Captured during rapid transmission across the population (*movement-speed* = 0.4). The monitors record **102 Vaccinated**, **319 Healthy**, **153 Infected**, **31 Recovered**, **184 Total Infections**, and **154 Peak Infected**. The spatial view shows a dense concentration of red infected agents and emerging cyan recovered agents spreading from the left region, while the *Epidemic Curve* plots the downward trajectory of healthy agents (green line) alongside rising infected (red) and recovered (blue) counts.

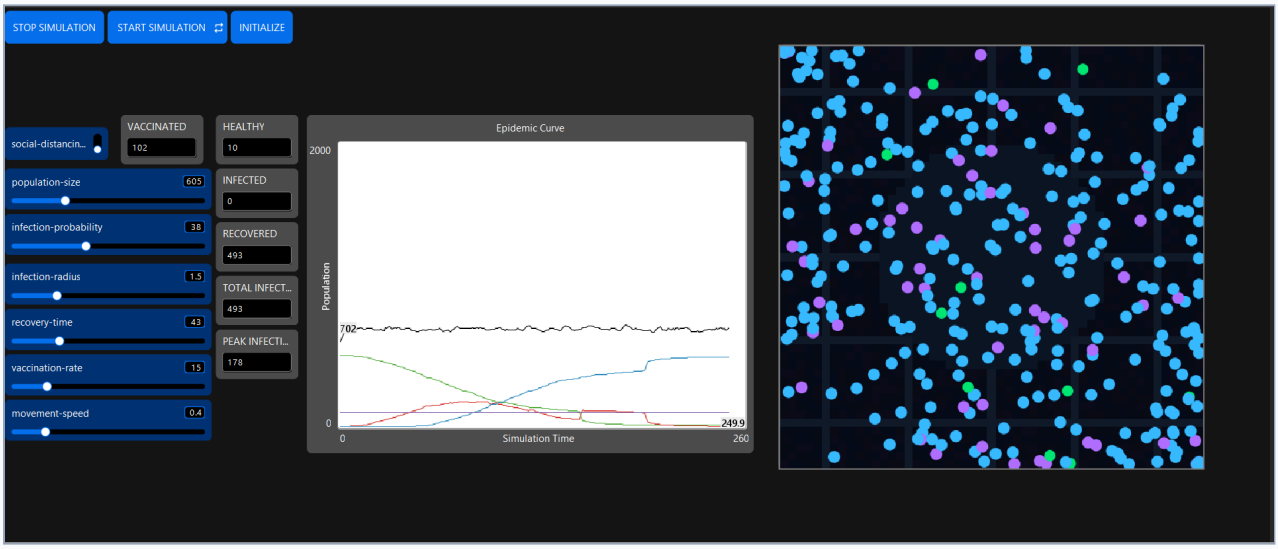


Figure 3: Later / Post-Outbreak Stage of the Simulation. Captured after the outbreak has run its course (around simulation time 249.9–260). The monitors record **102 Vaccinated**, **10 Healthy**, **0 Infected**, **493 Recovered**, **493 Total Infections**, and **178 Peak Infected**. No red infected agents remain in the simulation space, which is now populated predominantly by cyan recovered agents, purple vaccinated agents, and 10 remaining green healthy agents. The *Epidemic Curve* displays the complete rise and fall of the red infection wave and the plateau of the blue recovery curve.